WORKED SOLUTION

STEP III 2008, Question 5

Pure • Chebyshev's polynomial

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q5

The key idea

The quantity $\mathrm{T}_n^2 - \mathrm{T}_{n-1}\mathrm{T}_{n+1}$ is an *invariant* of the recurrence: when we step n up by one, the difference collapses because the recurrence lets us replace $\mathrm{T}_{n+1} + \mathrm{T}_{n-1}$ by $2x\mathrm{T}_n$. When the invariant is zero, consecutive ratios are all equal, so the sequence is geometric. Substituting a geometric sequence into the recurrence then pins down the common ratio as a root of a quadratic.

Solution

Throughout, x is fixed and we write T_n for $\mathrm{T}_n(x)$ and f for $\mathrm{f}(x)$. The recurrence $(*)$ says

$$\mathrm{T}_{n+1} + \mathrm{T}_{n-1} = 2x\,\mathrm{T}_n \qquad (n \geq 1).$$

The invariant

Let $D_n = \mathrm{T}_n^2 - \mathrm{T}_{n-1}\mathrm{T}_{n+1}$ for $n \geq 1$. We prove $D_n = f$ for all $n \geq 1$ by induction.

Base case. $D_1 = \mathrm{T}_1^2 - \mathrm{T}_0\mathrm{T}_2$, which is f by definition.

Inductive step. Suppose $D_k = f$ for some $k \geq 1$. Then

$$\begin{aligned} D_{k+1} - D_k &= \mathrm{T}_{k+1}^2 - \mathrm{T}_k\mathrm{T}_{k+2} - \mathrm{T}_k^2 + \mathrm{T}_{k-1}\mathrm{T}_{k+1} \\ &= \mathrm{T}_{k+1}(\mathrm{T}_{k+1} + \mathrm{T}_{k-1}) - \mathrm{T}_k(\mathrm{T}_{k+2} + \mathrm{T}_k) \\ &= \mathrm{T}_{k+1} \cdot 2x\mathrm{T}_k - \mathrm{T}_k \cdot 2x\mathrm{T}_{k+1} \\ &= 0, \end{aligned}$$

where in the third line we used $(*)$ with $n = k$ for the first bracket and with $n = k+1$ for the second (both allowed, as $k \geq 1$). Hence $D_{k+1} = D_k = f$, and by induction

$$\mathrm{T}_n^2 - \mathrm{T}_{n-1}\mathrm{T}_{n+1} = f \quad \text{for all } n \geq 1,$$

as required.

The case $f \equiv 0$

First let us see what $f = 0$ says about $r = \mathrm{T}_1/\mathrm{T}_0$. By $(*)$ with $n = 1$, $\mathrm{T}_2 = 2x\mathrm{T}_1 - \mathrm{T}_0$, so

$$f = \mathrm{T}_1^2 - \mathrm{T}_0(2x\mathrm{T}_1 - \mathrm{T}_0) = \mathrm{T}_0^2(r^2 - 2xr + 1),$$

on writing $\mathrm{T}_1 = r\mathrm{T}_0$. So, as $\mathrm{T}_0 \neq 0$,

$$f = 0 \iff r^2 = 2xr - 1. \qquad (\dagger)$$

> **Ask yourself** The invariant says $\mathrm{T}_n^2 = \mathrm{T}_{n-1}\mathrm{T}_{n+1}$, so the ratios are equal. Why not just divide?

Dividing by T_{n-1} needs it to be nonzero, which takes extra argument. It is cleaner to guess the answer from the "equal ratios" idea and then prove it with the recurrence itself, using $(\dagger)$.

Claim. $\mathrm{T}_n = \mathrm{T}_0\, r^n$ for all $n \geq 0$.

Proof. We use strong induction (two consecutive cases at a time). The claim holds for $n = 0$ trivially and for $n = 1$ because $\mathrm{T}_1 = r\mathrm{T}_0$ by the definition of r. Suppose it holds for $n = k - 1$ and $n = k$, where $k \geq 1$. Then by $(*)$,

$$\begin{aligned} \mathrm{T}_{k+1} &= 2x\mathrm{T}_k - \mathrm{T}_{k-1} = 2x\mathrm{T}_0 r^k - \mathrm{T}_0 r^{k-1} \\ &= \mathrm{T}_0 r^{k-1}(2xr - 1) = \mathrm{T}_0 r^{k-1} \cdot r^2 = \mathrm{T}_0 r^{k+1}, \end{aligned}$$

using $(\dagger)$ in the fourth step. So the claim holds for $n = k + 1$, and by induction for all $n \geq 0$:

$$\boxed{\mathrm{T}_n(x) = \mathrm{T}_0(x)\,(\mathrm{r}(x))^n}$$

The possible forms of $\mathrm{r}(x)$

By $(\dagger)$, r satisfies $r^2 - 2xr + 1 = 0$, so by the quadratic formula

$$\mathrm{r}(x) = x + \sqrt{x^2 - 1} \qquad \text{or} \qquad \mathrm{r}(x) = x - \sqrt{x^2 - 1}.$$

Conversely, either choice satisfies $(\dagger)$, so $f \equiv 0$ and the sequence $\mathrm{T}_0 r^n$ really does obey $(*)$. The two roots have product 1, so they are reciprocals of each other. For $|x| < 1$ they are complex, $x \pm \mathrm{i}\sqrt{1 - x^2}$; writing $x = \cos\theta$, they become $\mathrm{e}^{\pm \mathrm{i}\theta}$, and $\mathrm{T}_n = \mathrm{T}_0 \mathrm{e}^{\pm \mathrm{i}n\theta}$.

Checks

- Chebyshev polynomials $\mathrm{T}_0 = 1$, $\mathrm{T}_1 = x$, $\mathrm{T}_2 = 2x^2 - 1$ give $f = x^2 - (2x^2 - 1) = 1 - x^2$, and indeed $\cos^2 n\theta - \cos(n-1)\theta\cos(n+1)\theta = \sin^2\theta$ for every n. With $x = \frac{3}{7}$ the first fourteen values of D_n all equal $\frac{40}{49}$.
- Take $r = 2$, so $x = \frac{1}{2}(2 + \frac{1}{2}) = \frac{5}{4}$, and $\mathrm{T}_0 = 1$, $\mathrm{T}_1 = 2$. The recurrence gives $\mathrm{T}_2 = \frac{5}{2} \cdot 2 - 1 = 4$ and $\mathrm{T}_3 = \frac{5}{2} \cdot 4 - 2 = 8$: powers of 2, as claimed. Also $x - \sqrt{x^2 - 1} = \frac{5}{4} - \frac{3}{4} = \frac{1}{2}$ is the other root.

Common slips

- Applying $(*)$ with $n = 0$: the recurrence starts at $n = 1$, so the base case must be D_1, not D_0 (which is not even defined).
- Dividing $\mathrm{T}_n^2 = \mathrm{T}_{n-1}\mathrm{T}_{n+1}$ by T_{n-1} without checking it is nonzero.

Beyond STEP

A linear recurrence with constant coefficients has solutions $\alpha r_1^n + \beta r_2^n$, where r_1, r_2 are the roots of its characteristic equation, here $r^2 - 2xr + 1 = 0$. The condition $f = 0$ picks out exactly the cases where only one geometric term survives; in general f is a multiple of $\alpha\beta$. The invariant D_n is a discrete cousin of the Wronskian for second-order differential equations, which is constant for $y'' + q(x)y = 0$. Boyce and DiPrima, *Elementary Differential Equations*, develops the Wronskian, and Siklos, *Advanced Problems in Mathematics*, has more STEP-style recurrence work.

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